D.1 Baseline: End-to-End VLM Visual Reasoning

Listing A.1: VLM Visual Reasoning Prompt

```
SYSTEM_PROMPT = """
You are an expert visual analyst, working on Bongard Logo problems containing synthetic images with
    abstract shapes. You are presented with {n + m + 1} images:
   - The first {n} images are labeled 'cat_2' (positive examples).
   - The next {m} images are labeled 'cat_1' (negative examples).
   - The last image is the 'Test Image'.

   Your task is to analyze the visual features of all images and then classify the 'Test Image'
       based on the features it shares with 'cat_1' or 'cat_2'.

   **Your Task:**

   1. **Analyze Positive and Negative Images:**
       * Carefully compare the positive (cat_2) and negative (cat_1) images.
       * **Describe the visual characteristics of *each* category (cat_1 and cat_2) in detail.**
           Focus on shapes, lines, orientations, spatial relationships, and quantities. Break down
           complex shapes into simpler components. Describe shapes as if explaining them to someone
           who can't see them.
       * The goal is to implicitly understand the pattern distinguishing cat_2 from cat_1, even if
           you don't explicitly state it as a single rule.

   2. **Analyze the Test Image:**
       * Describe the test image's visual features in detail, using the same level of detail as in
           step 1.

   3. **Classify the Test Image:**
       * Based on your analysis of the positive, negative, and test images, determine whether the
           test image belongs to cat_2 (positive) or cat_1 (negative).
       * **Provide a clear justification for your classification.** Explain *why* the test image's
           features align more closely with the positive examples or the negative examples,
           referencing your detailed descriptions from steps 1 and 2.

   **Output Format:**

   * **Analysis of Positive and Negative Examples:** (Detailed descriptions of cat_1 and cat_2
       characteristics)
   * **Test Image Analysis:** (Detailed description of the test image's features)
   * **Test Image Classification:** (Either "cat_2 (Positive)" or "cat_1 (Negative)", followed by
       your justification)

   **Begin Analysis:**
   **Output Format (JSON):**
   }
       "Test Image": "(Detailed description of the test image's features)",
       "Rule": "(List of features identified as consistently present in cat_2 category)",
       "Analysis": "(Detailed descriptions of cat_1 and cat_2 characteristics)",
       "Conclusion": "(cat_1 or cat_2)"
   }

   Respond in the JSON format provided, and avoid unnecessary explanations or text outside of the
       JSON structure.

   **Begin Analysis of All Images:**
"""
```

D.2 Componential-Grammatical (C-G): Action Programs

Listing A.2: C-G Action Program Prompt

```
SYSTEM_PROMPT = """
**Objective:** Solve a Bongard-style visual reasoning puzzle by identifying the rule that
    distinguishes a "positive" set of programs from a "negative" set.

       **Input Context:**
```

```
        - You will receive human-readable, step-by-step instructions that describe how to construct a set of abstract shapes. These are called "action descriptions".
        - You will NOT be shown any images. Your task is to reason directly from these textual descriptions.

        **Your Core Task:**
        1. Analyze the provided "action descriptions" for the positive, negative, and test sets.
        2. Infer the final geometric properties of the shapes from these descriptions.
        3. Identify the abstract rule that defines the positive set.
        4. **Crucially, your reasoning must focus on the final geometric properties of the shapes, not the specific steps used to construct them.**

        Here is the overall concept behind the positive samples: {concept}

        **Your Task and Required Output:**
        Respond **only** with a single JSON object. Do not include any text or explanations outside of the JSON structure.

        The JSON object must have the following keys:
        }
            "Analysis": "A brief analysis of the distinguishing characteristics you observed in the 'pos' samples that make them distinct from 'neg' samples.",
            "Rule": "A clear and concise rule that defines the positive samples.",
            "Test Image": "A summary of the test sample's properties, as inferred from its description, and how it fits or doesn't fit the derived rule.",
            "Conclusion": "Your final categorization of the test image. The value must be either 'pos' or 'neg'."
        }
"""
```

D.3 Concept-Conditioned C–G

We add a single declarative sentence specifying the target concept, keeping the rest of the prompt identical across experiments to isolate the effect of explicit concept conditioning. Concept-Conditioned C–G uses the same prompt as in Listing A.2, with one additional line inserted before the "Your Task and Required Output" block:

Here is the overall concept behind the positive samples: {*concept*}

D.4 Natural Language Procedures (Ablation)

Listing A.3: Natural Language Action Description Prompt

```
SYSTEM_PROMPT = """
**Objective:** Solve a Bongard-style visual reasoning puzzle by identifying the rule that distinguishes a "positive" set of programs from a "negative" set.

        **Input Context:**
        - You will receive human-readable, step-by-step instructions that describe how to construct a set of abstract shapes. These are called "action descriptions".
        - You will NOT be shown any images. Your task is to reason directly from these textual descriptions.

        **Your Core Task:**
        1. Analyze the provided "action descriptions" for the positive, negative, and test sets.
        2. Infer the final geometric properties of the shapes from these descriptions.
        3. Identify the abstract rule that defines the positive set.
        4. **Crucially, your reasoning must focus on the final geometric properties of the shapes, not the specific steps used to construct them.**

        **Your Task and Required Output:**
        Respond **only** with a single JSON object. Do not include any text or explanations outside of the JSON structure.

        The JSON object must have the following keys:
        {
            "Analysis": "A brief analysis of the distinguishing characteristics you observed in the 'pos' samples that make them distinct from 'neg' samples.",
```

```
            "Rule": "A clear and concise rule that defines the positive samples.",
            "Test Image": "A summary of the test sample's properties, as inferred from its
                description, and how it fits or doesn't fit the derived rule.",
            "Conclusion": "Your final categorization of the test image. The value must be either
                'pos' or 'neg'."
        }
"""
```

D.5 Minimal-Context Baseline (Ablation)

Listing A.4: Minimal-Context Prompt

```
SYSTEM_PROMPT = """
**Your Task and Required Output:**
        Analyze the provided samples and respond *only* with a single JSON object. Do not include
            any additional text, explanations, or markdown formatting outside of the JSON structure.

        The JSON object must have the following keys:
        }
            "Analysis": "A brief analysis of the distinguishing characteristics you observed in the
                'pos' samples that make them distinct from 'neg' samples.",
            "Rule": "A clear and concise rule that defines the positive samples.",
            "Test Image": "A summary of the test image's properties as they relate to the derived
                rule.",
            "Conclusion": "Your final categorization of the test image. The value must be either
                'pos' or 'neg'."
        }
"""
```

D.6 Grounded Componential–Grammatical (Ablation)

In the Grounded C–G setting, models receive the same symbolic inputs as in the main C–G conditions, but the *query* also includes its rendered image. The goal is to test whether a single visual anchor, paired with otherwise symbolic input, provides additional benefits once an explicit visual language is available.

Concretely:

- For Gemini models, we call the official API with a text prompt and a PNG image object as a second content element.
- For open VLMs served via Ollama (e.g., Gemma3, LLaVA-LLaMA3, Qwen2.5-VL), we send a system message containing the text prompt and attach a single image encoded as base64.

In both cases, only the query image is passed; positive and negative examples are provided in symbolic form (action programs), and the model must output a JSON object with an analysis, an induced rule, a description of the test image, and a binary conclusion (pos/neg).

Grounded base (visual-only baseline). For the purely visual "base" context, the system prompt is:

Listing A.5: Minimal-Context Prompt

```
SYSTEM_PROMPT = """
    **Your Task and Required Output:**
    Analyze the provided samples and respond *only* with a single JSON object. Do not include any
        additional text, explanations, or markdown formatting outside of the JSON structure.

    The JSON object must have the following keys:
    }
        "Analysis": "A brief analysis of the distinguishing characteristics you observed in the
            'pos' samples that make them distinct from 'neg' samples.",
        "Rule": "A clear and concise rule that defines the positive samples.",
        "Test Image": "A summary of the test image's attributes, referencing the provided png
            image.",
        "Conclusion": "Your final categorization of the test image. The value must be either 'pos'
            or 'neg'."
    }
    """
```